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 - 3. As a student, how do you think cloud computing services could improve your school? Think about the ways that you turn in work, take exams, attend classes and events, or any other factor related to your school.
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 - 2. What are some examples of places that have different levels of access based on who you are? What are some things that people can use to prove that they have access to places?

Why are certain places restricted based on a person's access level?

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- 1. What might motivate someone to initiate a cyberattack against a company? What might attackers have to gain? Include an example of a company or a type of business and a kind of cyberattack it might be a victim of.
 - 2. Do you think there should be different security standards for the cloud based on the type of data that is being stored or processed? Why do you think that? Give an example. How do you think security differs between data stored in the cloud and data stored on premises?
 - 3. What character traits do you think a successful cloud security administrator would need? Why? Would this be a role that you would be interested in?

- Module 9: Monitoring the Cloud

- Present the concepts

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 - Amazon CloudWatch:
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- Answer the questions in the Question section

- 1. What tools do you use to stay organized and keep track of your life, work, and schedule? Why are these tools important? What kinds of tools would be helpful to monitor or keep track of your resources in the cloud?
 - 2. Have you ever missed or been late to an event you had scheduled, or forgotten an assignment? What happened? How might you have prevented the error? Do you think a similar error might happen when using cloud services with AWS? How might this be prevented?
 - 3. A cell phone company uses AWS to let users download mobile apps that let them print remotely from their devices. What data points do you think this company needs to keep track of in their cloud services? Why?

Module 6: Virtual Storage

Present the concepts

Module purpose

This module introduces Amazon Elastic Block Store (Amazon EBS) and provides a comparison between Amazon EBS and Amazon Simple Storage Service (Amazon S3). Students will explore the various storage tiers available and learn how to select the most appropriate storage solution for specific scenarios.

Module description

Participants will examine the key features and advantages of Amazon EBS, analyze case studies that showcase its real-world applications, and eventually create their own EBS volumes, attaching them to an Amazon Elastic Compute Cloud (Amazon EC2) instance.

Technology terminology

- Amazon Elastic Block Store (Amazon EBS)

Cloud-based storage designed for Amazon EC2 instances, functioning like an attached storage drive.>

- Amazon Elastic Compute Cloud (Amazon EC2)

A web service that provides scalable, secure compute resources in the cloud, acting like a virtual computer rental.

- Hard disk drive (HDD)

A slower storage device that stores data on a spinning disk.

- Input/Output Operations Per Second (IOPS)

A metric used to measure the speed and performance of storage devices like HDDs and SSDs.

- Solid state drive (SSD)

A high-speed storage device that uses flash memory instead of a spinning disk.

Background and misconceptions

Amazon EBS vs. Amazon S3: Key Differences

Amazon EBS (Elastic Block Store) provides block storage for EC2 instances, offering benefits like data availability, persistence, encryption, and snapshots. EBS is structured in fixed-length blocks, similar to file systems like NTFS or FAT, enabling fast access. It can only be attached to a single EC2 instance and has a more limited storage capacity compared to S3. Entire EBS volumes are encrypted.

Amazon S3 (Simple Storage Service) is an object storage system accessible via HTTP protocols, where data is stored as objects with metadata, requiring full reads or writes of objects. S3 can be accessed by multiple EC2 instances and typically experiences more latency when writing data compared to EBS. S3 provides various

storage classes such as S3 Standard, S3 Standard-IA, S3 One Zone-IA, S3 Intelligent-Tiering, S3 Glacier, and S3 Glacier Deep Archive. Objects in S3 are encrypted using server-side encryption (SSE).

The choice between EBS and S3 depends on application-specific requirements, including cost, performance, and throughput.

Answer the questions in the Question section

1. In your opinion, how has cloud computing impacted the way that society interacts with technology? Is it a positive or negative impact overall? Why?

In my view, cloud computing has drastically changed the way society interacts with technology, making it more accessible, scalable, and collaborative. Key positive impacts include:

- **Accessibility and Convenience:** Cloud computing allows access to data, applications, and services from anywhere with an internet connection, facilitating remote work, online learning, and global collaboration.
- **Cost Efficiency and Scalability:** It reduces the need for expensive hardware, helping businesses, especially startups, to scale without large upfront infrastructure investments.
- **Innovation and Agility:** On-demand computing power and storage have enabled rapid innovation, allowing developers to test, deploy, and iterate on applications faster.
- **Collaboration:** Cloud tools like Google Drive and Microsoft 365 enable real-time teamwork, breaking down geographic barriers.

However, there are downsides:

- **Security and Privacy:** Storing data in the cloud poses risks like data breaches and unauthorized access, requiring trust in third-party providers.
- **Dependence on Internet and Providers:** Cloud computing relies heavily on stable internet connections and service providers, making outages disruptive.

Overall, I believe cloud computing has had a largely positive impact, with its benefits outweighing the challenges when proper security measures are in place.

2. In an election, it is important to make sure that the vote count is accurate. Do you know how votes are actually counted? In some areas, election officials manually read each ballot and add up the number of votes in each race. Other areas have computerized voting systems that transmit vote totals to the central counting facility. Computerized voting is faster than counting ballots by hand, but some people argue that it is risky because it opens elections up to the possibility of hacking. Do you think that it is a good idea to use cloud services to protect and count votes? Why or why not?

Using cloud services to count and protect votes could offer significant benefits, such as speed and scalability, but it also carries risks. Cloud systems can count large amounts of votes quickly, providing faster results than manual counting. This can reduce delays and streamline the electoral process.

However, security is a major concern. Cloud systems are vulnerable to hacking and unauthorized access, which could undermine election integrity. Robust security measures, like encryption and multi-factor authentication, would be essential to protect voting data.

In conclusion, while cloud services could improve efficiency in elections, their security risks need to be thoroughly addressed to build public trust and prevent potential breaches.

3. As a student, how do you think cloud computing services could improve your school? Think about the ways that you turn in work, take exams, attend classes and events, or any other factor related to your school.

As a student, I believe cloud computing could significantly improve the school experience in several ways:

- **Easy Access to Resources:** Cloud storage enables students to access course materials and assignments from any location with internet access.
- **Collaboration Tools:** Platforms like Google Workspace and Microsoft 365 support real-time group collaboration, making project work more efficient.
- **Submission and Feedback:** Cloud-based assignment submissions streamline the process, making it easier for teachers to provide feedback and track progress.
- **Online Exams and Quizzes:** Cloud systems can facilitate secure, flexible online assessments and automatically grade quizzes for instant feedback.
- **Event Management:** Cloud services help organize events, manage RSVPs, and enhance communication between students and staff.
- **Learning Management Systems (LMS):** A cloud-hosted LMS can centralize all course-related information, helping students stay organized.
- **Backup and Recovery:** Cloud storage ensures assignments and important documents are protected and easily recoverable in case of device failure.

Overall, cloud computing can enhance collaboration, streamline processes, and provide flexibility, improving the overall student experience.

Module 7: Security I

Present the concepts

Module purpose

This module provides an overview of cloud security, focusing on AWS Identity and Access Management (IAM). Key topics include best practices, roles, users, policies, and security groups.

Module description

In this module, you will:

- Take notes on important information presented.
- Participate in discussions or role-plays to explore the social impacts of cloud security.
- Identify and resolve vulnerabilities in a web server.

Technology terminology

AWS Identity and Access Management (IAM):

A service to securely manage access to AWS resources by creating and controlling users, groups, roles, and permissions.

Role:

A set of permissions that define actions users or services can perform, providing temporary access to resources.

User:

An entity representing a person or application interacting with AWS, with specific permissions to access AWS resources.

Security Group:

A virtual firewall controlling traffic to AWS resources like EC2 instances.

Policy:

A JSON document that defines permissions for a user, group, or role, specifying allowed or denied actions.

Amazon Inspector:

A service that automatically assesses applications for vulnerabilities to improve security and compliance.

Group:

A collection of IAM users that can be managed together, with permissions applying to all members.

Root User:

The initial account with full access to AWS resources, recommended for use only in specific scenarios.

Credential:

Information like access keys, passwords, or MFA tokens that authenticate a user and provide access to AWS resources.

Enable Multi-Factor Authentication (MFA):

A security feature that requires a second factor (like a mobile device) in addition to a password.

JavaScript Object Notation (JSON):

A format for representing structured data, commonly used in IAM policies to define permissions.

Multi-Factor Authentication (MFA):

A security method requiring two or more verification factors for access to AWS resources.

Background and misconceptions

Security is critical for managing sensitive data in the cloud, such as banking or medical records. IAM helps control who can access AWS resources and what they can do. In organizations, multiple people across various departments may manage AWS accounts, making adherence to IAM best practices essential for maintaining a secure cloud environment.

AWS Identities: Key IAM components include roles, identities, and groups, governed by policies:

- **Root User:** The root user, which has full administrator access, should be reserved for tasks like account changes. Everyday tasks should be performed using an administrator account instead.
- **IAM User:** A user represents an individual accessing AWS services and is typically assigned to a group with appropriate permissions.
- **Group:** A group simplifies permission management by applying policies to all users within the group. For instance, an "Admins" group can automatically assign administrative permissions to its members.
- **IAM Roles:** Roles are temporary sets of permissions, assumed by users or services as needed, enhancing security by avoiding permanent credentials.
- **Policies:** Permissions are defined by policies stored as JSON documents, with best practices suggesting that policies be assigned to groups for efficient permission management.

Answer the questions in the Question section

1. What three things do you own that are most valuable to you? How do you secure each one? How can you secure something in multiple ways? How do you determine how secure something needs to be?

Three valuable things and how they are secured: *Personal Data (e.g., digital files, photos):*

- **Encryption:** Protect sensitive files by using encryption software, making them inaccessible without a password.
- **Backups:** Regularly store copies of data on secure cloud services and external hard drives.
- **Access Controls:** Use strong, unique passwords and enable multi-factor authentication (MFA) for services that store sensitive information.

Financial Information (e.g., bank accounts, investment details):

- **Strong Passwords:** Use complex and regularly updated passwords for financial accounts.
- **MFA:** Enable multi-factor authentication for extra security.
- **Monitoring:** Regularly check account activity for any unusual transactions and set alerts for suspicious activities.
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Valuable Physical Items (e.g., jewelry, electronics):

- **Safe Storage:** Store valuables in a secure safe or locked space.
- **Insurance:** Insure high-value items for protection against theft or damage.
- **Home Security System:** Install security measures like cameras and alarms to protect against theft.

To secure something in multiple ways, combine both physical and digital security measures, such as storing data with encryption and keeping physical backups. The security level needed depends on the item's value, sensitivity, potential threat, and any legal requirements.

2.What are some examples of places that have different levels of access based on who you are? What are some things that people can use to prove that they have access to places? Why are certain places restricted based on a person's access level?

Examples of places with varying levels of access:

- Offices and Workplaces: Employees may access common areas, but sensitive locations like server rooms may be restricted to higher-level staff.
- Hospitals: Doctors, nurses, and admin staff have different access levels based on their roles.
- Educational Institutions: Students access classrooms and libraries, but administrative areas are restricted to faculty.
- Government Buildings: Secure areas may be limited to authorized personnel.
- Events and Venues: VIP sections are accessible only to certain ticket holders.

Proof of access can include:

- ID cards, key fobs, access cards, passwords, PINs, or biometric scans.

Places are restricted to safeguard sensitive data, ensure safety, and meet legal standards for privacy and security.

3.Have you or someone you know ever had something stolen or broken into? How did it feel? Did it change how safe you felt you or your things were? How so? Did it change how you managed your security? How so?

Feelings after experiencing a theft or break-in:

- Violation and Vulnerability: A deep sense of violation often arises when personal spaces are intruded upon.
- Fear and Anxiety: People may feel less secure, leading to heightened fear.

This experience can change how safe a person feels, making them more aware of security risks and sometimes causing paranoia. In response, victims often improve security by installing better locks, alarms, or cameras, becoming more vigilant, or engaging with neighbors on safety measures. The incident can significantly influence how people manage their security in the future.

Module 8: Security II

Present the concepts

Module purpose

This module focuses on cloud security, emphasizing the differences between AWS Shield and AWS WAF. It covers Distributed Denial of Service (DDoS) attacks, as well as Amazon Inspector and AWS Artifact.

Module description

- Scenario Analysis: Review AWS Cloud scenarios to assess security measures.
- Best Practices Evaluation: Determine if security best practices are being followed.
- Recommendations: Provide solutions to fix any security lapses.

Technology terminology

AWS Shield:

A managed DDoS protection service for AWS applications. Shield Standard: Automatic, free protection for all AWS users. Shield Advanced: Offers enhanced protection with real-time attack visibility.

AWS WAF (Web Application Firewall):

Protects applications from web-based exploits. Users can define rules to filter and manage web traffic.

Distributed Denial of Service (DDoS):

A cyber attack that overloads services with traffic from multiple sources, causing downtime and disruptions.

Amazon Inspector:

An automated security assessment service that scans AWS applications for vulnerabilities and provides detailed reports.

AWS Artifact:

Provides access to AWS compliance reports and documentation to help manage compliance and security requirements.

Background and misconceptions

Cloud security involves four key areas:

- Data Protection: Safeguarding stored and processed information.
- Permissions: Controlling access to resources and data.
- Infrastructure: Securing the hardware and systems running cloud services.
- Assessment: Evaluating security by inspecting data, permissions, and infrastructure.

Key AWS Services:

- AWS Shield: Protects against DDoS attacks.
 - Standard: Free, automatic DDoS protection.
 - Advanced: Paid service with enhanced mitigation and cost protection.
- AWS WAF: Monitors and filters web traffic to protect against common exploits.
- Amazon Inspector: Evaluates AWS services for vulnerabilities and compliance, providing detailed findings.
- AWS Artifact: A central resource for compliance-related information, including AWS compliance standards for handling sensitive data.

Answer the questions in the Question section

1.What might motivate someone to initiate a cyberattack against a company? What might attackers have to gain? Include an example of a company or a type of business and a kind of cyberattack it might be a victim of.

- Financial Gain: Attackers may steal sensitive data (e.g., credit card information) for fraud or selling on the dark web. Example: A retail company could face a data breach where customer payment information is stolen.
- Revenge or Disgruntlement: Former employees or individuals may attack a company out of anger. Example: A tech firm's ex-employee might sabotage by deleting critical data or disrupting services.
- Political or Ideological Reasons: Hacktivists may attack organizations to push a political agenda. Example: A government agency could face a DDoS attack from a group protesting certain policies.
- Corporate Espionage: Competitors may steal trade secrets or proprietary data for a business advantage. Example: A pharmaceutical company could be targeted for its R&D data on a new drug.
- Challenge or Reputation: Some hackers attack for the thrill or to prove their skills. Example: A gaming company might face a DDoS attack as hackers seek notoriety by disrupting services.

Example Scenario: A financial institution could be hit by a ransomware attack, where attackers encrypt data and demand a ransom. This can severely disrupt operations, leading to financial losses and reputational damage.

2.Do you think there should be different security standards for the cloud based on the type of data that is being stored or processed? Why do you think that? Give an example. How do you think security differs between data stored in the cloud and data stored on premises?

Yes, different security standards should apply depending on the data type, as varying data levels of risk and sensitivity require tailored protection.

Reasons:

- Data Sensitivity: Personal health information (PHI) and financial records need stricter controls than less sensitive data.
- Regulatory Compliance: Industries like healthcare and finance have strict regulations like HIPAA and PCI DSS.
- Potential Impact of Breaches: Sensitive data breaches can cause severe financial, legal, and reputational damage.

Example: A healthcare provider storing patient data in the cloud must follow HIPAA regulations, using encryption, access control, and auditing. A company storing general marketing data may not require the same strict security measures.

Cloud vs. On-Premises Security:

- Cloud: Security is shared between the provider and the organization. Cloud providers secure infrastructure, while customers secure their applications and data.

- On-Premises: Organizations have full control over security but bear the responsibility for maintaining and updating it.
- Scalability: Cloud security can scale quickly with changing needs, while on-premises systems require additional resources and hardware to scale.

3.What character traits do you think a successful cloud security administrator would need? Why? Would this be a role that you would be interested in?

- Attention to Detail: Essential for spotting vulnerabilities and ensuring all security protocols are correctly implemented.
- Problem-Solving Skills: Necessary for analyzing complex issues and creating effective solutions.
- Adaptability: The cloud security landscape is ever-changing, so administrators must be adaptable and continually learning.
- Communication Skills: Important for explaining security policies to non-technical stakeholders and collaborating with teams.
- Integrity and Trustworthiness: Due to the sensitive data handled, administrators need to be highly ethical and trustworthy.
- Analytical Thinking: Recognizing patterns and analyzing data to detect threats early is crucial.

Interest in the Role: Yes, I find the role of a cloud security administrator intriguing, as it combines technical and strategic skills while playing a vital role in cybersecurity. The dynamic nature of the job, along with the opportunity to learn continuously, makes it highly appealing.

Module 9: Monitoring the Cloud

Present the concepts

Module purpose

The purpose of this module is to introduce AWS monitoring tools, specifically AWS Config, AWS CloudTrail, and Amazon CloudWatch.

Module description

Effective tracking of activities in cloud operations is essential due to the costs associated with each action. AWS provides tools to monitor, log, and report service usage.

In this module, you will:

- Explore AWS monitoring tools.
- Set up alerts using Amazon CloudWatch.
- Monitor AWS CloudTrail log files.
- Discuss the applications of CloudTrail and CloudWatch.
- Analyze scenarios to determine the best services to use.

This hands-on approach enhances your understanding of AWS monitoring capabilities and their importance in managing cloud resources effectively.

Technology terminology

Amazon CloudWatch:

Monitors AWS resources and applications in real-time.

AWS CloudTrail:

Logs every action in your AWS account for security and auditing.

AWS Config:

Assesses and evaluates the configurations of AWS resources.

Amazon Simple Notification Service (SNS):

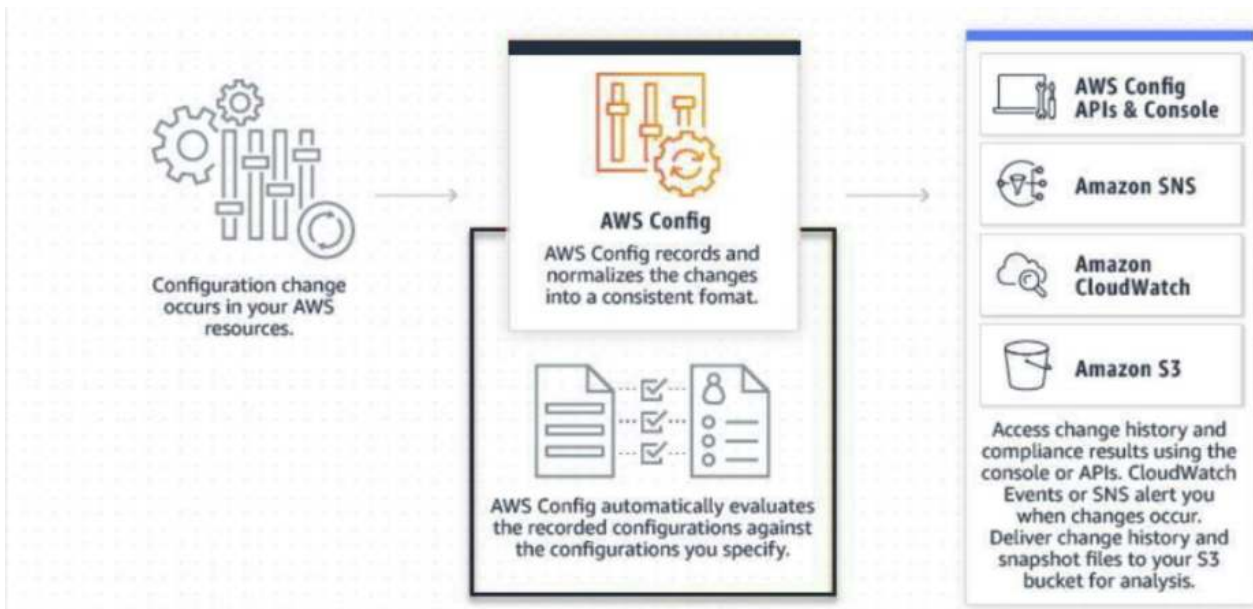
Sends notifications via texts, emails, and messages, enabling communication between cloud services and clients.

Background and misconceptions

AWS offers interconnected services for effective monitoring and management of cloud resources, which is crucial for businesses operating multiple AWS accounts.

- Amazon CloudWatch: Monitors resources and applications to ensure they run efficiently, helping track usage and performance for budget management.
- AWS CloudTrail: Logs all actions in an AWS account, providing a detailed history of user activities for security audits and compliance.
- AWS Config: Continuously monitors and records resource configurations, automating evaluations against desired settings to ensure compliance.
- Amazon SNS: Sends notifications based on events and alerts, facilitating timely information dissemination regarding resource changes.

Together, these tools create a robust framework for monitoring cloud operations, ensuring security, compliance, and optimal resource management.



AWS Config is a service that monitors and records configuration changes within your AWS environment. It automatically compares these changes against your desired state to identify deviations and non-compliance.

Key features and benefits:

- Continuous monitoring: Records all configuration changes.
- Automatic evaluation: Compares configurations against desired state.
- Compliance monitoring: Ensures adherence to predefined rules.
- Auditing: Provides a historical record of changes.
- Troubleshooting: Helps identify the root cause of issues.
- Improved security: Detects unauthorized modifications.
- Increased compliance: Meets regulatory requirements and best practices.
- Enhanced visibility: Provides deeper insights into AWS resources.
- Reduced risk: Proactively identifies and addresses potential problems.

By using AWS Config, you can improve security, ensure compliance, gain visibility, and reduce risk in your AWS environment.

Answer the questions in the Question section

1. What tools do you use to stay organized and keep track of your life, work, and schedule? Why are these tools important? What kinds of tools would be helpful to monitor or keep track of your resources in the cloud?.

To stay organized, I use:

- Digital Calendars (e.g., Google Calendar): For scheduling and reminders.
- Task Management Apps (e.g., Todoist, Trello): To create to-do lists and track project progress.
- Note-taking Apps (e.g., Evernote, Notion): For organizing ideas and important documents.
- Time Tracking Tools (e.g., Toggl): To monitor time spent on tasks and optimize workflow.

These tools are crucial for efficient responsibility management, stress reduction, and meeting deadlines.

For cloud resource monitoring, useful tools include:

- AWS CloudWatch: For performance and usage metrics.
- AWS CloudTrail: For logging actions for security and auditing.
- AWS Config: To ensure compliance of resource configurations.
- Cost Management Tools (e.g., AWS Cost Explorer): For tracking spending and optimizing costs.

2. Have you ever missed or been late to an event you had scheduled, or forgotten an assignment? What happened? How might you have prevented the error? Do you think a similar error might happen when using cloud services with AWS? How might this be prevented?

I've missed deadlines before due to not setting reminders. To prevent this, I could have set multiple reminders and regularly checked my calendar.

In cloud services like AWS, similar errors can occur if tasks are overlooked. To mitigate this risk:

- Automated Alerts: Set notifications via AWS CloudWatch or Amazon SNS.
- Regular Audits: Use AWS Config and CloudTrail for routine checks.
- Documentation: Maintain thorough records for reference.

3. A cell phone company uses AWS to let users download mobile apps that let them print remotely from their devices. What data points do you think this company needs to keep track of in their cloud services? Why?

For a cell phone company using AWS for remote printing, essential data points include:

- User Activity Logs: To monitor app usage and optimize features.
- Print Job Data: To manage resource allocation and costs.
- Error and Performance Metrics: For troubleshooting and improving user experience.
- User Feedback and Ratings: To guide app enhancements.
- Billing and Usage Statistics: For revenue management in subscription models.
- Security Logs: To protect user privacy and ensure compliance.
- Resource Utilization Metrics: To optimize performance and manage costs.

Tracking these data points enhances service delivery, maintains security, and improves operational efficiency.